

12.7 Percent Increase and Decrease

Lesson Introduction

 Benchmark	MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.		 Learning Target	I can find percent increase or percent decrease.	
 Benchmark Coherence	Building On		Working Toward		
	MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.		N/A		
 Essential Questions	How can you model percent change problems with a tape diagram and double number lines? How do you calculate percent increase and percent decrease?				
 Lesson Narrative	Students solve a variety of real-world percent change problems. Initially, students use tape diagrams and double number lines to model the problems. Then, students are introduced to the percent of change formula and apply that to additional problems in the lesson.				
 Component Summary	12.7.1 Warm-Up (~5 min.)	Use a percentage to determine the fewest number of people surveyed (<i>SE p. 247</i>)	12.7.4 Collaboration (10-15 min.)	Solve problems involving percent change (<i>SE p. 250</i>)	
	12.7.2 Guided Instruction (10 min.)	Learn how to find the percent of change using models (<i>SE p. 248</i>)	12.7.5 Your Turn! (5 min.)	Additional practice solving problems involving percent change (<i>SE p. 252</i>)	
	12.7.3 Try It (10 min.)	Use models to represent percent change (<i>SE p. 249</i>)	12.7.6 Wrap-Up (~5 min.)	Error analysis of percent change (<i>SE p. 253</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> Percent of change <u>Additional Terms:</u> N/A		N/A		Consider creating and displaying an anchor chart with instructions, examples, and visuals for students to reference.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle to understand why they can add the percent increase to 100% or subtract the percent decrease from 100%. - Students may reverse the order of the values in the percent of change formula.	- If students struggle with percent change formulas, it may help to walk them through the process to get to $\frac{\text{difference}}{\text{original}}$. This includes students recognizing that the percent of change will be positive for an increase and negative for a decrease. - Consider allowing students to use calculators for the 12.7.4 Collaboration, 12.7.5 Your Turn!, and 12.7.6 Wrap-Up. Allowing students to use calculators during the lesson will help students focus on discovery and the process of calculating percent of change. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with operations with rational numbers. For example, students should not need a calculator in the components prior to 12.7.4 Collaboration.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Take Turns	MA.K12.MTR.4.1: Discussion and Reflection
	For the 12.7.4 Collaboration, have students take turns explaining how they solved questions 1 and 2. Partners can provide feedback and compare their work. These discussions will help students gain confidence and understanding.	The 12.7.6 Wrap-Up provides the opportunity for students to analyze the percent of change error in a sample of student work. Through identifying and correcting mistakes in a sample of student work, students will be less likely to make mistakes themselves.
Lesson Notes:		

12.7.1 Warm-Up: Interpreting Percentages

Component Overview: Students interpret percentages in a real-world problem by considering what the fewest number of people surveyed was.



12.7.1 Warm-Up: Interpreting Percentages

1. If exactly 93.6% of people completed a survey, what is the fewest number of people that were surveyed? Consider: How could it be determined if only one person could have completed it? Can partial people complete the survey?

125 is the fewest number of people that could have been surveyed. If one person completed the survey, then the percentage would have to be 100%. You cannot have a fractional part of a person complete a survey. For exactly 93.6% of people to complete a survey, the amount of people surveyed must be at least 125.



This component is scaffolded such that parts could be released to students at this point in the unit. Consider which questions may need more modeling or elaboration and which would be best for students to complete with a partner.

2. Explain your answer using mathematics.

I rewrote the percentage as a fraction and then used equivalent fractions to eliminate the decimal by putting the percentage over 1000 since I can not have a decimal amount of a person. Then, I reduced it to lowest terms.

$$\frac{93.6}{100} \times \frac{10}{10} = \frac{936}{1000} \div \frac{8}{8} = \frac{117}{125}$$

12.7.2 Guided Instruction: Percent of Change

Component Overview: Students develop an understanding of percent change by modeling problems with tape diagrams and double number lines. The models provide students with visual entry points to the concept of percent change.



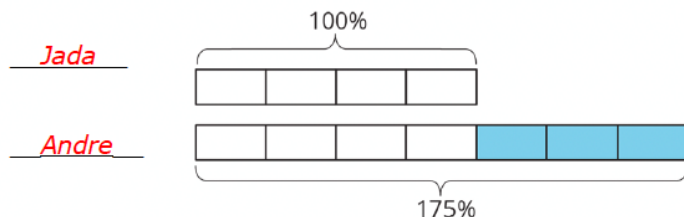
12.7.2 Guided Instruction: Percent of Change

Percentages can be used to make comparisons from one quantity to another.

- It takes Andre $\frac{3}{4}$ times longer than it took Jada to get to school.

The diagram models this in terms of percentages.

- Label each line of the tape diagram with the name of the student who is represented.



- Complete the statements.

The time it takes Andre to get to school is 75% ☐ more ☐ less

than the time it takes Jada. The diagram shows Andre's

travel time is ☐ 75% ☐ 175% of Jada's travel time.

- Let x represent Jada's time. Write an expression to model how long it takes Andre to get to school.
 $1.75x$
- If it takes Jada 8 minutes to get to school, calculate how long it takes Andre to get to school.
 14 minutes



Consider asking students the following:

- Does it take Andre more or less time to get to school than Jada?
- What percentage is equivalent to $\frac{3}{4}$?

12.7.2 Guided Instruction: Percent of Change (continued)

The terms “percent increase” and “percent decrease” are used to describe a specific **percent of change**.



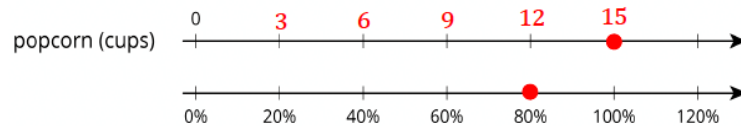
A **percent of change** is the difference between a final value and an initial value, expressed as a percentage of the initial value.



If time allows, consider having students share different strategies they used, or describe where they got stuck if they weren't able to determine an answer.

A double number line can also be used to model percent increase or percent decrease.

2. A movie theater decreased the size of popcorn bags by 20%. The old bags held 15 cups of popcorn.



- On the top number line, plot the 15 cups of popcorn as the initial value of 100%.
Shown on top number line.
- On the second line, plot 20% less than the initial 100%.
Shown on second number line.
- Use the double number line to find the number of cups in the smaller bag.
12 cups
- Discuss with your partner how the double number line is helpful for determining the impact of a 20% decrease in the size of the popcorn bag.
Answers will vary. Each segment on the number line represents 3 cups which is equivalent to 20%. A decrease of 20% shows that 3 cups need to be taken away from the initial 15 cups.



If the amount of popcorn is decreased by 20%, what percentage of the original size is the new size?

12.7.3 Try It: Percent of Change

Component Overview: Students practice using tape diagrams and double number lines to model percent change problems. Consider having students discuss their work with a partner after initially completing the questions independently.



12.7.3 Try It: Percent of Change

1. Explain which diagram, A or B, matches each situation.

Diagram A	Diagram B
Last year 100% This year 25%	Last year 100% This year 25%



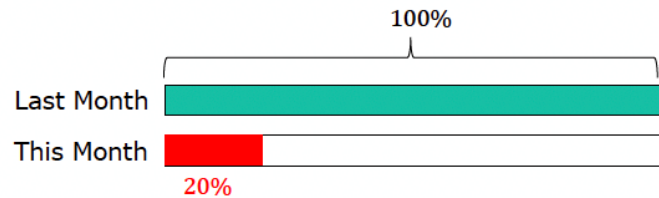
Ask students how they can distinguish between the two diagrams.

- How can you tell if the question pertains to percent increase based on the diagram?
- How can you tell if the question pertains to percent decrease based on the diagram?

Situation	Matching Diagram	Reason
Compared to last year's strawberry harvest, this year's increased 25%.	A	Diagram A matches because this year is 25% more than last year.
This year's blueberry harvest is 75% of last year's.	B	Diagram B matches because this year is 25% less than last year.
Compared to last year, this year's peach harvest decreased by 25%.	B	Diagram B matches because this year is 25% less than last year.
This year's plum harvest is 125% of last year's plum harvest.	A	Diagram A matches because this year is 25% more than last year.

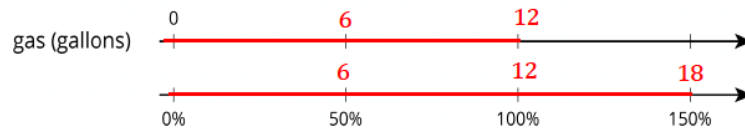
12.7.3 Try It: Percent of Change (continued)

2. The number of mosquitoes decreased by 80% from last month to this month. Create a tape diagram of this month to represent the situation.



If there were 100 mosquitoes last month, how many were there this month?

3. The gas tank in Robert's truck holds 12 gallons. The gas tank in his mom's truck holds 50% more gas. Using the double number lines, determine how many gallons the gas tank of Robert's mom's truck holds.



The gas tank in Robert's mom's truck holds 18 gallons.

12.7.4 Collaboration: Calculating Percent Increase or Decrease

Component Overview: Students collaboratively solve percent increase and decrease problems. Students are introduced to the percent change formula and then use it to solve a problem.



12.7.4 Collaboration: Calculating Percent Increase or Decrease

Percent increase and decrease can also be found through calculations.

Complete the following with your partner.

1. A new cereal box contains “20% More” cereal. The original cereal boxes contained 500 grams of cereal.



- a. Explain how to determine how much cereal is in the new box.
Find what 20% of 500 is. Then add it to 500. This will be 120% of 500.

- b. How much cereal comes in the new box?

$$500 + 0.2(500) = 600 \text{ grams}$$

2. The price of a shirt at ClothesMart is \$18.50, but there is a store coupon that lowers the price by 20%.

- a. Work with your partner to determine how much the shirt is with the coupon.

$$18.5 - 0.2(18.5) = \$14.80$$



Take Turns

Ask students to take turns solving each problem and then explaining their solution. This routine ensures each and every student has the opportunity to articulate their mathematical thinking and critique the reasoning of others.



There are multiple strategies for solving the problems in this component. Encourage students to use strategies that make sense for them as they build fluency in answering questions 1 and 2.

12.7.4 Collaboration: Calculating Percent Increase or Decrease (continued)

The table shows two different ways to complete this problem, along with corresponding answers.

Approach 1	Approach 2
The cost of the shirt with the coupon will be 80% of the original cost.	The new cost of the shirt is what's remaining after the 20% has been deducted from the original cost.
New cost = $0.8(18.50)$	New cost = $18.50 - 0.2(18.50)$



Algebra Talk

Ask students to compare the numeric expressions and discuss what is the same and different. How is the 20% discount represented in each?

- b. Explain whether or not these equations are equivalent.

These equations are equivalent because both can be used to find the new cost correctly. In Approach 1, the discount is removed from the percentage to find 80% of the original cost. In Approach 2, the price of the discount is first calculated and then subtracted from the original cost.

- c. Did your answer to Part A match the solution(s) of the equations in the table?

Yes



$$\text{Percent of Change} = 100 \times \frac{\text{Final Value} - \text{Initial Value}}{\text{Initial Value}}$$

3. The original cost of pants at ClothesMart is \$30.00, and after a store coupon is applied, the cost is only \$25.50.

- a. Determine the percent of change by filling in the blanks of the percent of change ratio for the situation.

$$\frac{\boxed{25.50} - \boxed{30}}{\text{Initial}} = \frac{\boxed{25.50} - \boxed{30}}{\boxed{30}} = \frac{\boxed{-4.5}}{30} = \boxed{-15} \%$$

12.7.4 Collaboration: Calculating Percent Increase or Decrease (continued)

- b. Explain whether the percent of change represents a percent increase or decrease based on its value.

This is a percent decrease because a discount decreases the price. You can also tell because the resulting change is a negative percentage.

- c. Complete the statements.

When calculating the percent of change using the ratio,

a positive value represents a percent

- ☐ increase,
☐ decrease,

and a negative value represents a percent

- ☐ increase.
☐ decrease.

In this context, it makes sense that the percent change is a

percent

- ☐ increase
☐ decrease

because the change is the result of a

discount.



Students may struggle to understand the series of boxes in question 3. If needed, walk through the steps with students or draw the problem on the board. Label each box with the information it should contain such as “initial cost,” “cost after coupon,” “difference,” and “percent change.” Then, have students fill the numbers in the boxes and use them to verbally explain the problem.

Be sure to provide students the opportunity to try it with their partner before jumping in to model. Consider also sharing strategies you observe on the board and asking students to compare them and determine errors.

12.7.5 Your Turn!

Component Overview: Students practice solving problems involving percent change.



12.7.5 Your Turn!

1. A school had 1,200 students last year and only 1,080 students this year. What was the percentage decrease in the number of students?

$$\frac{1080 - 1200}{1200} = \frac{-120}{1200} = -10\%$$

There was a 10% decrease in the number of students.



Consider connecting the context in these and other problems in the lesson to familiar or local situations to help students make connections to the problems. For example, perhaps there was a change in the number of students at their school or there is a package advertising 20% more that could be brought in for reference. Student motivation can be ignited if the problems involve people, places, and things that they are interested in.

2. A juice box has 20% more juice in its new packaging. The original packaging held 12 fluid ounces. How much juice does the new packaging hold?

$$1.2(12) = 14.4 \text{ fluid ounces}$$

12.7.6 Wrap-Up: Error Analysis

Component Overview: Students identify and correct an error in a percent change calculation. Consider using this information from this formative assessment to determine which students need remediation before the next lesson.



12.7.6 Wrap-Up: Error Analysis

- The number of campers who attend a local summer camp has increased from 1,938 to 2,128. The camp director calculates the percent of increase for an upcoming presentation. Her work is shown.

$$\begin{aligned} 2,128 - 1,938 &= 190 \\ 190 \div 2,128 &= 0.089 \\ 0.089 &= 8.9\% \end{aligned}$$

Explain the camp director's mistake.

The answer should be about 9.8% because $\frac{2,128-1,938}{1,938}$ is about 0.098. The camp director divided the change in number of campers by the new number instead of the original.



Students who are done early can also be asked to find the correct answer.